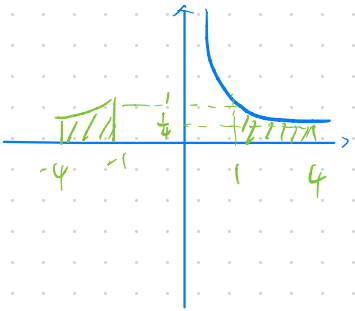
Exercises 6.3 5, 7, 11, 13, 15, 17, 19, 25(a), 39, 41, 43.

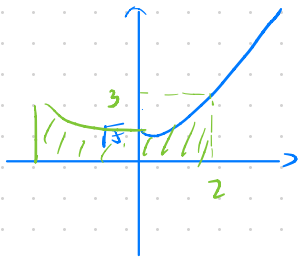
9–14 Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the y-axis.

11. $y = 1/x, \quad y = 0, \quad x = 1, \quad x = 4$



$$A = \int_1^4 2\pi r h \, dx$$
$$= \int_1^4 2\pi \cdot x \cdot \frac{1}{x} \cdot dx$$
$$= 2\pi \int_1^4 dx = 2\pi [x]_1^4 = 2\pi [4 - 1] = 6\pi$$

13. $y = \sqrt{5 + x^2}, \quad y = 0, \quad x = 0, \quad x = 2$



$$A = \int_0^2 2\pi r h \, dx$$
$$= \int_0^2 2\pi \cdot x \cdot \sqrt{5+x^2} \, dx$$
$$= 2\pi \int_0^2 x \sqrt{5+x^2} \, dx$$
$$= 2\pi \int_5^9 x \sqrt{u} \cdot \frac{1}{2x} \, du$$
$$= \pi \int_5^9 \sqrt{u} \, du = \pi \cdot \frac{2}{3} [u^{\frac{3}{2}}]_5^9$$
$$= \frac{2\pi}{3} [9^{\frac{3}{2}} - 5^{\frac{3}{2}}]$$
$$= 33.376$$

let $u = 5 + x^2$; $\frac{du}{dx} = 2x$

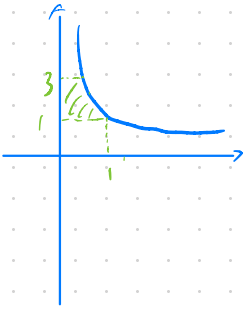
$dx = \frac{du}{2x}$

when $x = 2$, $u = 9$

$x = 0$, $u = 5$

15–20 Use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by the given curves about the x-axis.

15. $xy = 1, \quad x = 0, \quad y = 1, \quad y = 3$



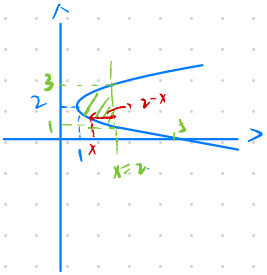
$A = \int_1^3 2\pi r h \, dy$

$= \int_1^3 2\pi \cdot y \cdot \frac{1}{y} \, dy$

$= 2\pi \int_1^3 dy = 2\pi [y]_1^3$

$= 2\pi \times 2 = 4\pi$

19. $x = 1 + (y - 2)^2, \quad x = 2$



$r = 2 - x$

$h = 2 - [1 + y^2 - 4y + 4]$

$= 2 - (y^2 - 4y + 3)$

$= -y^2 + 4y - 3$

$A = \int_1^3 2\pi r h \, dy$

$= 2\pi \int_1^3 y(-y^2 + 4y - 3) \, dy$

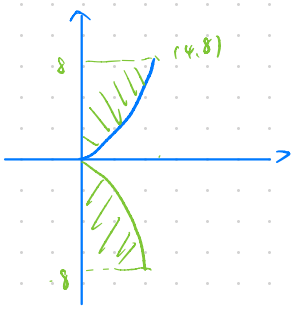
$= 2\pi \int_1^3 -y^3 + 4y^2 - 3y \, dy$

$= 2\pi \cdot \left[-\frac{1}{4}y^4 + \frac{4}{3}y^3 - \frac{3}{2}y^2\right]_1^3$

$= 2\pi \left[\left(-\frac{3^4}{4} + \frac{4}{3} \cdot 3^3 - \frac{3}{2} \times 3^2\right) - \left(-\frac{1}{4} + \frac{4}{3} - \frac{3}{2}\right)\right]$

$= \frac{16}{3}\pi$

17. $y = x^{3/2}, \quad y = 8, \quad x = 0$



$A = \int_0^8 2\pi r h \, dy$

$= \int_0^8 2\pi y \cdot y^{\frac{1}{3}} \, dy$

$= 2\pi \int_0^8 y^{\frac{4}{3}} \, dy$

$= 2\pi \cdot \frac{3}{7} \left[y^{\frac{7}{3}} \right]_0^8$

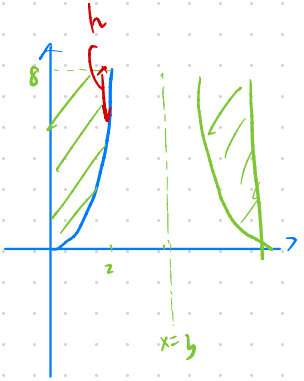
$y^2 = (x^{\frac{1}{2}})^2$

$y^{\frac{2}{3}} = x$

$2\pi \left[\frac{3}{7} y^{\frac{7}{3}} \right]_0^8$

25–30 Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the specified axis.

25. $y = x^3, \ y = 8, \ x = 0;$ about $x = 3$



$h = 8 - x^3$

$A = \int_0^2 2\pi r h \, dx$

$= \int_0^2 2\pi (3 - x)(8 - x^3) \, dx$

$= 2\pi \int_0^2 3x^3 - x^4 \, dx$

$= 2\pi \left[\frac{3}{4} x^4 - \frac{1}{5} x^5 \right]_0^2$

$=$

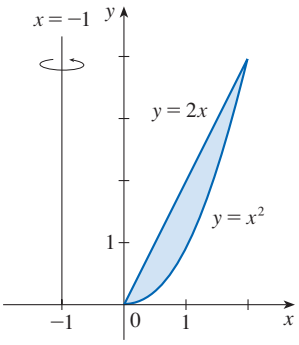


FIGURE 11

EXAMPLE 5 Figure 11 shows the region in the first quadrant bounded by the curves $y = x^2$ and $y = 2x$. A solid is formed by rotating the region about the line $x = -1$. Find the volume of the solid using (a) x as the variable of integration and (b) y as the variable of integration.

SOLUTION The solid is shown in Figure 12(a).

(a) To find the volume using x as the variable of integration, we draw the sample rectangle vertically, as in Figure 12(b). Rotating the region about the line $x = -1$ produces cylindrical shells, so the volume is

$$V = \int_0^2 2\pi(x + 1)(2x - x^2)\,dx = 2\pi \int_0^2 (x^2 + 2x - x^3)\,dx$$

$$= 2\pi\left[\frac{x^3}{3} + x^2 - \frac{x^4}{4}\right]_0^2 = \frac{16\pi}{3}$$

(b) To find the volume using y as the variable of integration, we draw the sample rectangle horizontally as in Figure 12(c). Rotating the region about the line produces washer-shaped cross-sections, so the volume is

$$V = \int_0^4 \left[\pi\left(\sqrt{y} + 1\right)^2 - \pi\left(\tfrac{1}{2}y + 1\right)^2\right] dy = \pi \int_0^4 \left(2\sqrt{y} - \tfrac{1}{4}y^2\right) dy$$

$$= \pi\left[\tfrac{4}{3}y^{3/2} - \tfrac{1}{12}y^3\right]_0^4 = \frac{16\pi}{3}$$

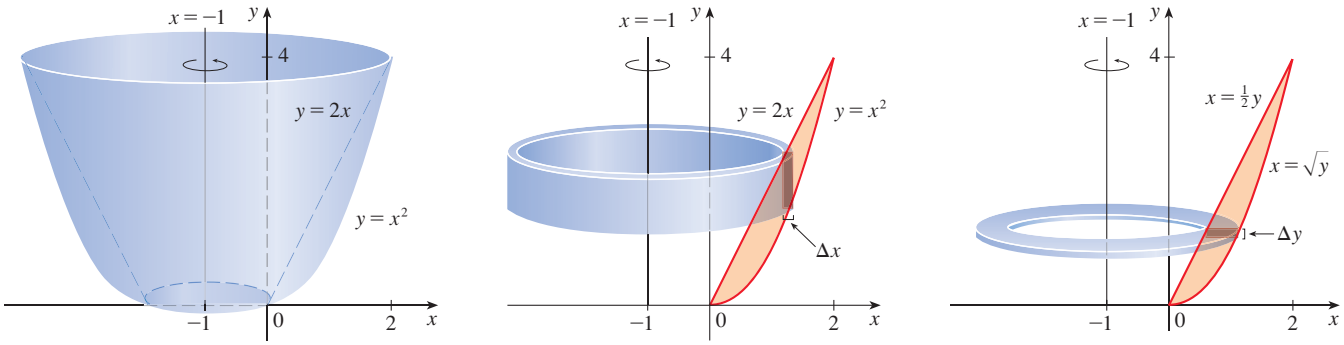
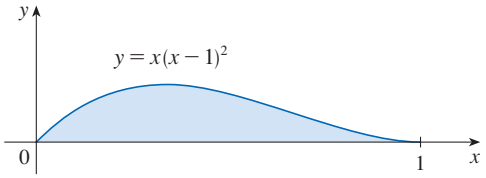


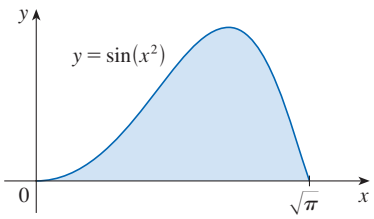
FIGURE 12

6.3 Exercises

1. Let S be the solid obtained by rotating the region shown in the figure about the y-axis. Explain why it is awkward to use the washer method to find the volume V of S. Sketch a typical approximating shell. What are its circumference and height? Use shells to find V.



2. Let S be the solid obtained by rotating the region shown in the figure about the y-axis. Sketch a typical cylindrical shell and find its circumference and height. Use shells to find the volume of S. Is this method preferable to using washers?

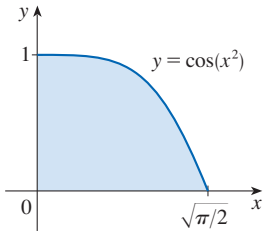


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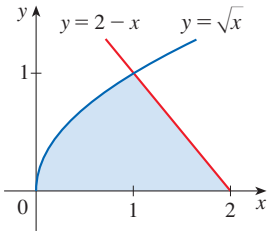
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3–4 A solid is obtained by rotating the shaded region about the specified line.
(a) Set up an integral using the method of cylindrical shells for the volume of the solid.
(b) Evaluate the integral to find the volume of the solid.

3. About the y-axis



4. About the x-axis



5–8 Set up, but do not evaluate, an integral for the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

5. $y = \ln x,\ y = 0,\ x = 2;$ about the y-axis

6. $y = x^3,\ y = 8,\ x = 0;$ about the x-axis

7. $y = \sin^{-1}x,\ y = \pi/2,\ x = 0;$ about $y = 3$

8. $y = 4x - x^2,\ y = x;$ about $x = 7$

9–14 Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the y-axis.

9. $y = \sqrt{x},\quad y = 0,\quad x = 4$

10. $y = x^3,\quad y = 0,\quad x = 1,\quad x = 2$

11. $y = 1/x,\quad y = 0,\quad x = 1,\quad x = 4$

12. $y = e^{-x^2},\quad y = 0,\quad x = 0,\quad x = 1$

13. $y = \sqrt{5 + x^2},\quad y = 0,\quad x = 0,\quad x = 2$

14. $y = 4x - x^2,\quad y = x$

15–20 Use the method of cylindrical shells to find the volume of the solid obtained by rotating the region bounded by the given curves about the x-axis.

15. $xy = 1,\quad x = 0,\quad y = 1,\quad y = 3$

16. $y = \sqrt{x},\quad x = 0,\quad y = 2$

17. $y = x^{3/2},\quad y = 8,\quad x = 0$

18. $x = -3y^2 + 12y - 9,\quad x = 0$

19. $x = 1 + (y - 2)^2,\quad x = 2$

20. $x + y = 4,\quad x = y^2 - 4y + 4$

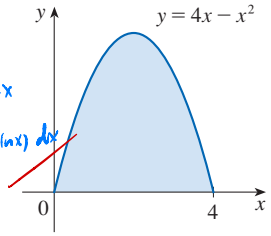
21–22 The region bounded by the given curves is rotated about the specified axis. Find the volume of the resulting solid using (a) x as the variable of integration and (b) y as the variable of integration.

21. $y = x^2,\ y = 8\sqrt{x};$ about the y-axis

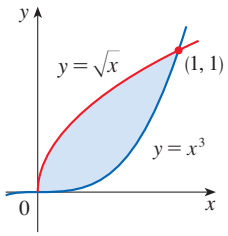
22. $y = x^3,\ y = 4x^2;$ about the x-axis

23–24 A solid is obtained by rotating the shaded region about the specified axis.
(a) Sketch the solid and a typical approximating cylindrical shell.
(b) Use the method of cylindrical shells to set up an integral for the volume of the solid.
(c) Evaluate the integral to find the volume.

23. About $x = -2$



24. About $y = -1$



25–30 Use the method of cylindrical shells to find the volume generated by rotating the region bounded by the given curves about the specified axis.

25. $y = x^3,\ y = 8,\ x = 0;$ about $x = 3$

26. $y = 4 - 2x,\ y = 0,\ x = 0;$ about $x = -1$

27. $y = 4x - x^2,\ y = 3;$ about $x = 1$

28. $y = \sqrt{x},\ x = 2y;$ about $x = 5$

29. $x = 2y^2,\ y \geqslant 0,\ x = 2;$ about $y = 2$

30. $x = 2y^2,\ x = y^2 + 1;$ about $y = -2$

31–36
(a) Set up an integral for the volume of the solid obtained by rotating the region bounded by the given curve about the specified axis.
T (b) Use a calculator or computer to evaluate the integral correct to five decimal places.

31. $y = xe^{-x},\ y = 0,\ x = 2;$ about the y-axis

32. $y = \tan x,\ y = 0,\ x = \pi/4;$ about $x = \pi/2$

33. $y = \cos^4 x,\ y = -\cos^4 x,\ -\pi/2 \leqslant x \leqslant \pi/2;$ about $x = \pi$

34. $y = x,\ y = 2x/(1 + x^3);$ about $x = -1$

35. $x = \sqrt{\sin y},\ 0 \leqslant y \leqslant \pi,\ x = 0;$ about $y = 4$

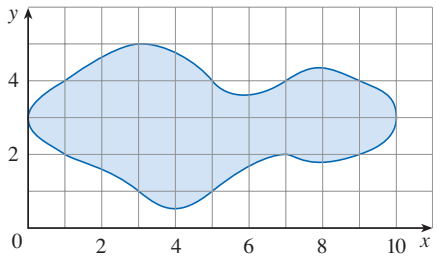
36. $x^2 - y^2 = 7,\ x = 4;$ about $y = 5$

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37. Use the Midpoint Rule with $n = 5$ to estimate the volume obtained by rotating about the y-axis the region under the curve $y = \sqrt{1 + x^3}$, $0 \leq x \leq 1$.

38. If the region shown in the figure is rotated about the y-axis to form a solid, use the Midpoint Rule with $n = 5$ to estimate the volume of the solid.



39–42 Each integral represents the volume of a solid. Describe the solid.

39. $\int_0^3 2\pi x^5\,dx$

40. $\int_1^3 2\pi y \ln y\,dy$

41. $2\pi \int_1^4 \dfrac{y+2}{y^2}\,dy$

42. $\int_0^1 2\pi(2-x)(3^x - 2^x)\,dx$

T **43–44** Use a graph to estimate the x-coordinates of the points of intersection of the given curves. Then use this information and a calculator or computer to estimate the volume of the solid obtained by rotating about the y-axis the region enclosed by these curves.

43. $y = x^2 - 2x, \quad y = \dfrac{x}{x^2+1}$

44. $y = e^{\sin x}, \quad y = x^2 - 4x + 5$

T **45–46** Use a computer algebra system to find the exact volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

45. $y = \sin^2 x,\ y = \sin^4 x,\ 0 \leq x \leq \pi$; about $x = \pi/2$

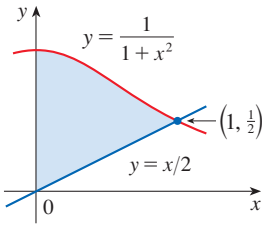
46. $y = x^3 \sin x,\ y = 0,\ 0 \leq x \leq \pi$; about $x = -1$

47–52 A solid is obtained by rotating the shaded region about the specified line.

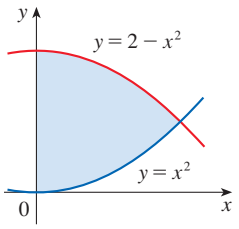
(a) Set up an integral using any method to find the volume of the solid.

(b) Evaluate the integral to find the volume of the solid.

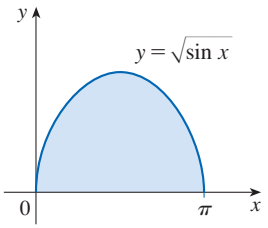
47. About the y-axis



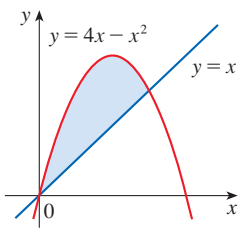
48. About the x-axis



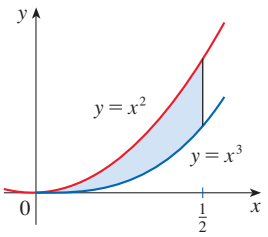
49. About the x-axis



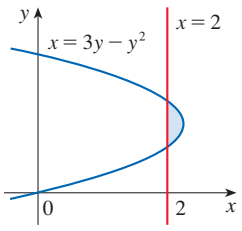
50. About the y-axis



51. About the line $x = -2$



52. About the line $y = 3$



53–59 The region bounded by the given curves is rotated about the specified axis. Find the volume of the resulting solid by any method.

53. $y = -x^2 + 6x - 8,\ y = 0$; about the y-axis

54. $y = -x^2 + 6x - 8,\ y = 0$; about the x-axis

55. $y^2 - x^2 = 1,\ y = 2$; about the x-axis

56. $y^2 - x^2 = 1,\ y = 2$; about the y-axis

57. $x^2 + (y-1)^2 = 1$; about the y-axis

58. $x = (y-3)^2,\ x = 4$; about $y = 1$

59. $x = (y-1)^2,\ x - y = 1$; about $x = -1$

60. Let T be the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$, and let V be the volume of the solid generated when T is rotated about the line $x = a$, where $a > 1$. Express a in terms of V.

61–63 Use cylindrical shells to find the volume of the solid.

61. A sphere of radius r

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